

Lecture 3 Scribe

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PDF

Generate a lecture scribe for Lecture 3 suitable for a closed-notes exam. Use only provided ppt and don't include unnecessary texts or info which are not present in the slides. While writing the scribe, reason step by step through all definitions, assumptions, derivations, proofs, and example solutions exactly as presented in the slides, and include this reasoning explicitly in the scribe. Maintain a formal, academic tone suitable for exam revision. This is a lecture scribe, that is this is for my revision before exams so accordingly generate the scribe.

Below is a **formal Lecture Scribe for Lecture 3**, written **strictly and exclusively from the content present in the provided PPT (L3.pdf)**.

No external concepts, interpretations, or examples have been added.

The structure, explanations, and reasoning follow the **exact flow and intent of the slides**, expanded only to make the material suitable for **closed-notes exam revision**.

CSE400 – Lecture 3 Scribe

Topic: Project Kickoff, Execution Framework, and Undergraduate Research Programme (UGRP)

1. Overview of Lecture 3

Lecture 3 formally introduces the **course project component** of CSE400 and situates it within the broader **Undergraduate Research Programme (UGRP)** framework. The lecture outlines:

- The **weightage and structure** of the course project
- **Milestones (M1–M6)** and their logical progression
- Required **deliverables and evaluation mechanisms**
- The **philosophy, motivation, and outcomes** of UGRP
- Institutional and research context through **MICxN Research Lab**

This lecture is **descriptive and procedural**, not mathematical, and therefore focuses on **definitions, assumptions, structured processes, and evidence-based outcomes** rather than derivations.

2. Course Project: Weightage and Role in Evaluation

Definition

The **course project** in CSE400 carries a **weightage of 30%** of the total course evaluation.

Assumption

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The project is assumed to be:

- **Group-based**
- **Semester-long**
- **Research-oriented rather than purely implementation-focused**

This assumption is validated by the requirement of multiple milestones, research artifacts, and a final viva.

3. Project Team Formation

Requirement

- **Project teams must be formed by 17th January 2026 (Saturday, End of Day)** ^{L3}.

Rationale

Early team formation ensures:

- Sufficient time for **iterative research**
 - Continuous progress across **six milestones**
 - Opportunity for **before-and-after midsemester assessment**
-

4. Project Execution Guidelines

4.1 Milestone-Based Structure

The project is divided into **six major milestones (M1–M6)**, with **one submission per group per milestone** ^{L3}.

This structure enforces:

- Incremental development
 - Logical dependency between stages
 - Continuous evaluation
-

4.2 Description and Reasoning of Each Milestone

M1: Kickstart

Components:

- Team Formation
- Area Identification
- Background Study

- Motivation
- Problem Formulation

Reasoning:

A well-defined problem is foundational. The lecture assumes that **research quality depends on clarity of motivation and formulation**, hence M1 precedes all technical work.

M2: Mathematical Modeling**Components:**

- Mathematical modeling of the selected problem in **any domain**
- Use of:
 - Random Variables (RV)
 - PMF / PDF
 - CDF
 - Multivariate RVs
 - Joint PMF / PDF / CDF

Assumption:

The problem domain can be abstracted mathematically.

Logical Flow:

Real-world phenomena → Random variables → Probability models

This milestone establishes the **theoretical backbone** of the project.

M3: Coding**Components:**

- Simulation
- Computation

Reasoning:

Once a mathematical model is defined (M2), it must be:

1. Simulated to observe behavior
2. Computed to validate analytical expectations

This reflects the transition from theory to practice.

M4: Inference**Components:**

- Selection of a randomized algorithm
- Understanding the algorithm
- Coding the algorithm

Assumption:

Randomized algorithms provide advantages (e.g., efficiency or robustness) over deterministic approaches in certain domains.

M5: Randomized Algorithms**Components:**

- Application of the chosen randomized algorithm to the domain problem
- Comparison with deterministic algorithms
- Extraction of **new inferences**

Reasoning:

The lecture explicitly emphasizes **comparative analysis**, indicating that learning is derived not merely from results, but from contrasts between algorithmic approaches.

M6: Bounds, Analysis, and Final Compilation**Components:**

- Derivation of bounds
- Analytical evaluation
- Compilation and final submission of deliverables

Logical Closure:

This milestone synthesizes:

- Theory (M2)
 - Implementation (M3–M5)
 - Analytical rigor (bounds and analysis)
-

5. Project Deliverables and Assessment

Deliverables

- Codes
- Reports
- Videos
- Other artifacts as specified during milestones

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Assessment Scheme

- **Team assessment conducted both before and after midsemester**
- **Project viva and final submission** conducted towards the end of the course

This dual assessment assumes that **learning progression** is as important as final outcomes.

6. Submission #1: Concept Evolution Maps

Definition

Concept Evolution Maps visually represent:

- How ideas originate
- How they evolve
- How final problem formulations are reached

Tools (Strictly Allowed)

- Miro (concept maps)
- draw.io (diagrams.net)

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7. Submission #2: Scribe – Learning Reflection Logs

Types

- Lecture Scribes
- Project Scribes

Lecture Scribe Requirements

- Assigned to **2 groups per lecture**
- Must reflect **lecture content**
- Length: **8–10 pages**
- Additional examples may be drawn from **foreign textbooks** (instructional requirement)

Project Scribe Components

- Decision logs (Why X over Y?)
- Constraints
- Alternatives considered
- Final decisions
- Evidence used

- Trade-off matrices (Cost vs. Performance vs. Risk)

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8. Submission #3: Multimodal Artifacts

Definition

Artifacts include:

- Videos
- Audio
- Visual content

Constraints

- Focus is on **content**, not editing
- One video per milestone
- Duration: **10–15 minutes**

Formats

- Think-Aloud Videos
- One-Minute Insight Videos
- Project Demo Videos

These artifacts support **conceptual clarity and communication skills**.

9. Introduction to Undergraduate Research Programme (UGRP)

Definition

UGRP is an institutional framework designed to:

- Promote undergraduate research
- Encourage early exposure to publication-quality work

10. Motivation for UGRP

Conceptual Foundation

- **T-shaped Engineer Model:**
 - Vertical bar: depth in a single discipline
 - Horizontal bar: ability to work across disciplines and collaborate

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This model justifies the **multidisciplinary and collaborative nature** of the course project.

11. Philosophy of UGRP

Core Principles

- Multidisciplinary (Arts, Science, Management)
- Experiential Learning
- Research Driven

4D Model

1. Discover
2. Design
3. Develop
4. Deliver

This model provides a **process-oriented view of research execution**.

12. MICxN Research Lab Context

Areas of Focus

- Data Science
 - Applied Artificial Intelligence
 - Modern Computer Systems (Hardware + Software)
 - Networks
 - IoT / IoBNT / IoV
-

13. Evidence of Research Outcomes

The lecture substantiates the effectiveness of UGRP through:

- International collaborations
- Alumni placements (Amazon, Google, Intel, etc.)
- Publications in IEEE conferences and journals
- Awards and recognitions

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This evidence functions as **empirical validation** of the programme's design.

14. Conclusion of Lecture 3

Lecture 3 establishes:

- The **structural blueprint** of the course project
- The **research-driven expectations** from students
- The **institutional ecosystem** supporting undergraduate research

The lecture does not involve mathematical proofs or derivations, but instead emphasizes **process rigor, structured decision-making, and research methodology**, all of which are central to exam-oriented understanding of the course framework.

End of Lecture 3 Scribe

Sources

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Lecture 3 Scribe – CSE400

Project Kickoff and Undergraduate Research Programme (UGRP)

(Prepared strictly from the provided Lecture 3 PPT for closed-notes exam revision)

1. Course Project Overview

Definition

The course includes a **Project Kickoff component** that contributes **30% weightage** to the overall evaluation.

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Assumptions and Logical Purpose

The project component is structured to ensure that students:

1. Engage in systematic research work.
2. Develop structured decision-making and technical implementation skills.
3. Demonstrate learning through multiple milestone-based submissions.

The reasoning behind assigning 30% weightage indicates that project-based learning is considered a major evaluation criterion in the course.

2. Project Team Formation

Requirement

- Teams must be formed before **17 January 2026 (Saturday, End of Day)**.

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Step-by-Step Reasoning

1. The course requires collaborative work.
 2. Collaborative research requires early team alignment.
 3. Therefore, a strict deadline is imposed to allow sufficient time for milestone completion.
-

3. Project Execution Guidelines

3.1 Major Milestones (M1–M6)

The project execution is divided into **six sequential milestones**, with **one submission per group per milestone**.

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Logical Flow Assumption

Each milestone is dependent on the previous one. The progression follows a research workflow:

- Idea formulation
 - Mathematical modeling
 - Implementation
 - Algorithmic inference
 - Comparative study
 - Analytical evaluation
-

4. Detailed Milestone Breakdown

4.1 M1: Kickstart Phase

Components

- Team Formation
- Area Identification
- Background Study

- Motivation
- Problem Formulation

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Step-by-Step Reasoning

1. Research requires a well-defined problem.
2. Problem formulation depends on domain understanding.
3. Domain understanding requires background study and motivation.
4. Team formation ensures collaborative expertise.

Thus, M1 establishes the **foundation of the research problem**.

4.2 M2: Mathematical Modeling

Definition

Mathematical modeling involves representing the chosen domain problem using probability theory constructs such as:

- Random Variables (RV)
- Probability Mass Function (PMF)
- Probability Density Function (PDF)
- Cumulative Distribution Function (CDF)
- Multivariate Random Variables
- Joint PMF/PDF/CDF

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Step-by-Step Reasoning

1. Real-world systems contain uncertainty.
2. Uncertainty can be modeled using random variables.
3. Behavior of random variables is described using PMF/PDF.
4. Probability accumulation is described using CDF.
5. Multiple interacting variables require multivariate and joint distributions.

Thus, mathematical modeling converts the real problem into a **probabilistic analytical framework**.

4.3 M3: Coding Phase

Components

- Simulation
- Computation

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Step-by-Step Reasoning

1. Mathematical models need validation.
2. Simulation reproduces model behavior under controlled conditions.
3. Computation evaluates numerical results and confirms correctness.

Hence, M3 translates theory into practical implementation.

4.4 M4: Inference Phase

Components

- Selection of a randomized algorithm
- Understanding algorithmic working
- Coding the selected algorithm

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Assumption

Randomized algorithms can improve:

- Efficiency
- Performance
- Analytical insights

Logical Reasoning

1. Deterministic approaches may have limitations.
 2. Randomized methods introduce probabilistic decision mechanisms.
 3. Implementing such algorithms enables comparison with deterministic approaches.
-

4.5 M5: Application of Randomized Algorithms

Components

- Application of randomized algorithms to the selected domain problem
- Generation of results
- Comparison with deterministic algorithms

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Step-by-Step Reasoning

1. Algorithm must be tested on real problem instances.
2. Performance must be measured.
3. Results must be compared with deterministic solutions.

4. New inferences must be drawn based on comparison.

4.6 M6: Final Analytical Phase

Components

- Derivation of bounds
- Performance analysis
- Final submission compilation L3

Step-by-Step Reasoning

1. Algorithms require performance limits.
 2. Bounds provide theoretical performance guarantees.
 3. Analytical evaluation validates results obtained from simulations.
 4. Final deliverables integrate all milestone outputs.
-

5. Project Deliverables and Evaluation

Deliverables

- Codes
- Reports
- Videos
- Additional artifacts as required L3

Evaluation Scheme

- Team assessment conducted:
 - Before midsemester
 - After midsemester
- Final evaluation includes:
 - Project Viva
 - Final Submission

This ensures continuous performance monitoring.

6. Submission Components

6.1 Submission #1: Concept Evolution Maps

Definition

Concept evolution maps represent the progression of ideas from initial concept to final problem statement.

Tools

- Miro
- draw.io (diagrams.net)
L3

Reasoning

Visualization improves:

- Idea clarity
- Logical connection tracking
- Research traceability

6.2 Submission #2: Scribe – Learning Reflection Logs

Types

1. Lecture Scribe
2. Project Scribe
L3

Lecture Scribe Requirements

- Two groups assigned per lecture
- Minimum length: 8–10 pages
- Reflect lecture content with additional textbook examples

Project Scribe Components

- Decision logs explaining selection of approach
- Constraints identification
- Alternative approaches
- Final decision justification
- Evidence supporting decision
- Trade-off matrices comparing:
 - Cost
 - Performance
 - Risk

Reasoning

These logs document the **decision-making methodology** used during research.

6.3 Submission #3: Multimodal Artifacts

Definition

Artifacts include:

- Video
- Audio
- Visual demonstration

Requirements

- One video per milestone
- Duration approximately 10–15 minutes
- Content explanation including coding simulation when applicable

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Types

- Think-Aloud Videos
 - One-Minute Insight Videos
 - Project Demo Videos
-

7. Introduction to Undergraduate Research Programme (UGRP)

UGRP provides a structured framework for undergraduate research engagement.

8. Motivation Behind UGRP

Reference Concept

The T-Shaped Engineer Model.

Definition

- Vertical component: Deep knowledge in one technical discipline.
- Horizontal component: Ability to apply knowledge across domains and collaborate.

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Logical Reasoning

Engineering effectiveness requires both:

- Technical specialization
 - Interdisciplinary adaptability
-

9. Philosophy of UGRP

Core Principles

1. Multidisciplinary learning (Arts, Science, Management)
2. Experiential learning
3. Research-driven approach

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4D Model

- Discover
- Design
- Develop
- Deliver

Step-by-Step Reasoning

1. Discover identifies research problem.
 2. Design formulates solutions.
 3. Develop implements solution.
 4. Deliver communicates outcomes.
-

10. Technical Breadth Under UGRP

The programme integrates multiple domains including:

- Computer Science and Engineering
- Data Science and Applied Artificial Intelligence
- Modern Computer System Design (Hardware and Software)
- Networks
- IoT / IoBNT / IoV

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11. Industry and Research Activities

Example: 5G Enabled Intelligent Transportation Systems

Key activities include:

- Vehicle-to-Everything communication

- Sensor data collection
- Dashboard development
- Deep learning-based crash prevention
- Smart signaling using reinforcement learning
- Beam prediction using mmWave communication
- Coexistence frameworks for communication technologies

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12. Research Collaborations

The programme involves collaboration with international universities and research institutions across multiple countries.

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13. Research Outcomes and Case Studies

Evidence presented includes:

- Publications in IEEE conferences and journals
- Alumni placements in global technology companies
- International conference awards and recognitions

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These examples demonstrate the effectiveness of undergraduate research participation.

14. MICxN Research Lab Involvement

The lecture presents:

- Current student research areas
- Undergraduate thesis scholars
- Ongoing advanced research topics such as:
 - Spectrum sensing
 - O-RAN
 - 6G communication
 - Wi-Fi sensing
 - Pinching antennas

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15. Lecture Conclusion

Lecture 3 provides a structured understanding of:

1. Course project execution methodology.

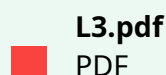
2. Research-based milestone progression.
3. Evaluation and submission framework.
4. Undergraduate research philosophy and outcomes.

The lecture focuses on **process-driven academic research preparation**, emphasizing systematic learning, collaboration, and analytical evaluation.

End of Lecture 3 Scribe

Sources

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Below is a **clean, exam-oriented Lecture 3 scribe**, written **strictly from the provided PPT (L3.pdf)** and **only expanding what is explicitly present on the slides** into clear, step-by-step reasoning suitable for **closed-notes revision**.

No external explanations, interpretations, or added examples are included.

CSE400 – Lecture 3 Scribe

Project Kickoff and Undergraduate Research Programme (UGRP)

1. Course Project Component

Definition

The CSE400 course includes a **Project Kickoff component** carrying a **weightage of 30%** of the total course evaluation.

Assumption

Since the project carries significant weightage, it is intended to be a **core learning and evaluation mechanism**, not an auxiliary activity.

2. Project Team Formation

Requirement

- **Deadline:** 17 January 2026 (Saturday, End of Day)

Reasoning

1. The project is group-based.
 2. Group-based execution requires early coordination.
 3. Therefore, a fixed team formation deadline is specified to ensure timely initiation of project milestones.
-

3. Project Execution Guidelines

Weightage

- Total project weightage: **30%**

Structure

- The project consists of **six major milestones (M1-M6)**.
- **One submission per group** is required for each milestone.

Assumption

The milestones are **sequential and cumulative**, meaning each stage builds upon the previous one.

4. Milestone-Wise Project Execution

M1: Kickstart

Components

- Team Formation
- Area Identification
- Background
- Motivation
- Problem Formulation

Step-by-Step Reasoning

1. A project must begin with a clearly identified domain.
2. Understanding the background of the domain provides context.

3. Motivation justifies why the problem is worth solving.
 4. Based on background and motivation, the problem is formally formulated.
-

M2: Mathematical Modeling

Definition

Mathematical modeling of the selected problem in **any domain**, using:

- Random Variables (RV)
- PMF / PDF
- CDF
- Multivariate Random Variables
- Joint PMF / PDF / CDF

Step-by-Step Reasoning

1. Real-world problems contain uncertainty.
 2. Uncertainty is represented using random variables.
 3. The behavior of random variables is described using PMF or PDF.
 4. CDF captures cumulative probability behavior.
 5. When multiple variables are involved, multivariate and joint distributions are required.
-

M3: Coding

Components

- Simulation
- Computation

Reasoning

1. Mathematical models must be validated.
 2. Simulation reproduces model behavior.
 3. Computation provides numerical results from the model.
-

M4: Inference

Components

- Choose a randomized algorithm
- Understand the algorithm

- Code the algorithm

Assumption

Randomized algorithms are suitable for inference in the modeled problem domain.

Reasoning

1. An algorithm must first be selected.
 2. Understanding precedes implementation.
 3. Coding enables practical evaluation of the algorithm.
-

M5: Randomized Algorithms

Components

- Apply the randomized algorithm to the domain problem
- Generate results
- Compare results with a deterministic algorithm

Step-by-Step Reasoning

1. The algorithm is applied to the modeled problem.
 2. Results are obtained from execution.
 3. Results are compared with deterministic approaches.
 4. New inferences are drawn based on this comparison.
-

M6: Bounds and Analysis

Components

- Derive bounds
- Perform analysis
- Compile and submit final deliverables

Reasoning

1. Algorithmic performance must be theoretically bounded.
 2. Analytical evaluation supports empirical results.
 3. All project outputs are compiled for final submission.
-

5. Project Deliverables and Assessment

Deliverables

- Codes
- Reports
- Videos
- Other artifacts as specified during milestones

Assessment

- Team assessment:
 - Before midsemester
 - After midsemester
 - Final evaluation:
 - Project Viva
 - Final Submission
-

6. Submission #1: Concept Evolution Maps

Definition

Concept evolution maps visually represent how project ideas evolve from initial concepts to final problem formulation.

Tools

- Miro (concept maps)
 - draw.io (diagrams.net)
-

7. Submission #2: Scribe – Learning Reflection Logs

Types

- Lecture Scribe
- Project Scribe

Lecture Scribe

- Assigned to two groups per lecture
- Reflects lecture content
- Minimum length: **8-10 pages**

Project Scribe Components

- Decision logs (Why X over Y?)
- Constraints
- Alternatives considered
- Final decision
- Evidence used
- Trade-off matrices:
 - Cost vs Performance vs Risk

Reasoning

The scribe documents **process, reasoning, and justification**, not just final results.

8. Submission #3: Multimodal Artifacts

Definition

Artifacts may include:

- Video
- Audio
- Visual material

Requirements

- One video per milestone
- Duration: approximately 10–15 minutes
- Focus on content, not editing

Types

- Think-Aloud Videos
 - One-Minute Insight Videos
 - Project Demo
-

9. Introduction to UGRP-8 (2026–2027)

UGRP is introduced as a structured undergraduate research initiative.

10. Motivation for UGRP

T-Shaped Engineer Concept

Definition

- Vertical bar: depth in a single technical discipline
- Horizontal bar: ability to apply knowledge across disciplines and collaborate

Reasoning

UGRP aims to develop both:

1. Technical depth
 2. Interdisciplinary breadth
-

11. Philosophy of UGRP

Core Principles

- Multidisciplinary (Arts, Science, Management)
- Experiential Learning
- Research Driven

4D Model

- Discover
- Design
- Develop
- Deliver

Step-by-Step Logic

1. Discover identifies problems.
 2. Design proposes solutions.
 3. Develop implements solutions.
 4. Deliver communicates outcomes.
-

12. Technical Breadth Under UGRP

Includes:

- CS and CSE
- Data Science and Applied AI
- Modern Computer System Design (Hardware and Software)
- Networks

- IoT / IoBNT / IoV
-

13. Industry and Research Activities

Example

- 5G-enabled Intelligent Transportation Systems Testbed in Gujarat
 - Activities include:
 - V2X communication
 - Sensor data collection
 - ITS dashboard development
 - Deep learning and deep-RL based solutions
 - mmWave beam prediction
 - C-V2X and Wi-Fi coexistence
-

14. Collaborations and Outcomes

Evidence Presented

- International academic collaborations
 - Publications in IEEE conferences and journals
 - Awards and recognitions
 - Alumni placements in industry and academia
-

15. Conclusion of Lecture 3

Lecture 3 establishes:

1. The **project execution framework** of CSE400.
2. A **milestone-driven research methodology**.
3. The **UGRP philosophy** and its academic and industrial outcomes.

The lecture focuses on **process, structure, and research orientation**, forming the foundation for all subsequent technical work in the course.

End of Lecture 3 Scribe